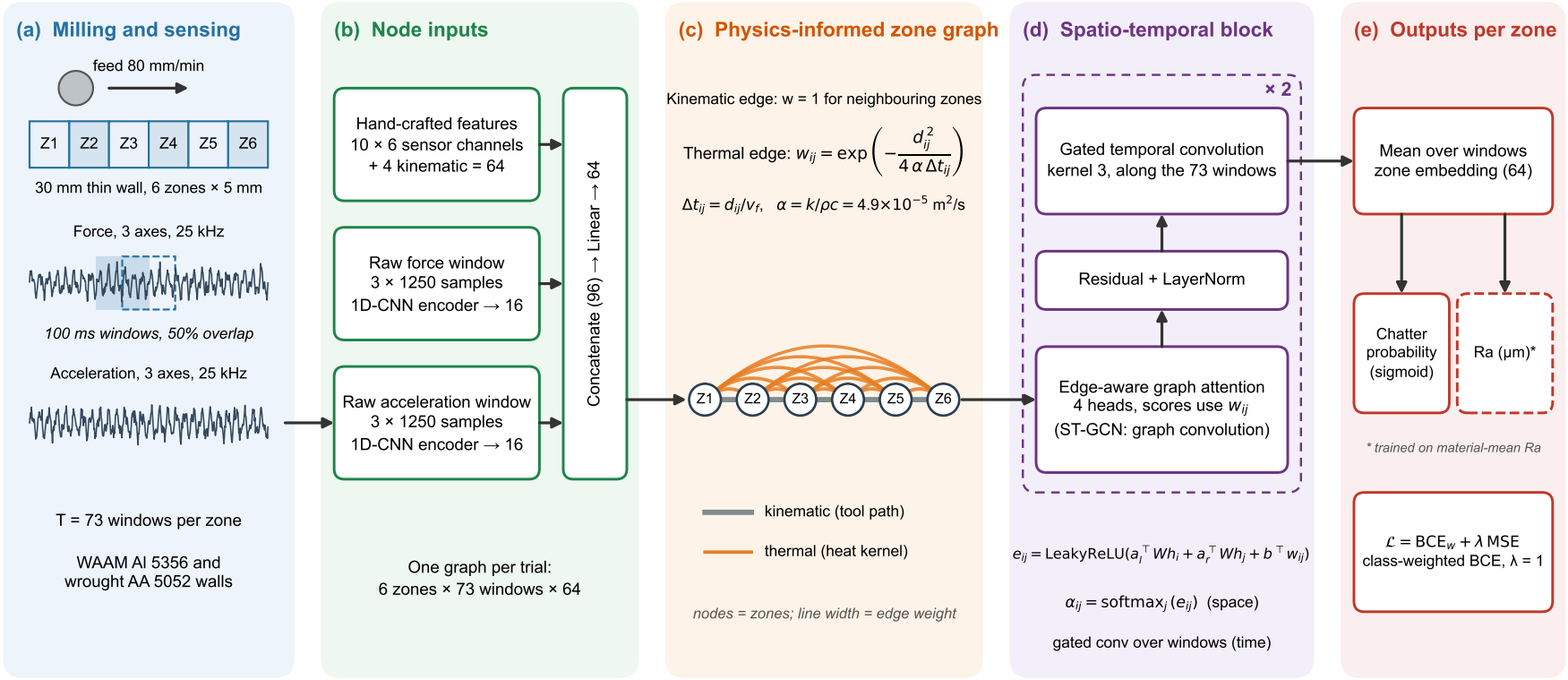

(a) Milling and sensing
feed 80 mm/min
Z1
Z2
Z3
Z4
Z5
Z6
30 mm thin wall, 6 zones × 5 mm
Force, 3 axes, 25 kHz
100 ms windows, 50% overlap
Acceleration, 3 axes, 25 kHz
T = 73 windows per zone
WAAM Al 5356 and
wrought AA 5052 walls
(b) Node inputs
Hand-crafted features
10 × 6 sensor channels
+ 4 kinematic = 64
Raw force window
3 × 1250 samples
1D-CNN encoder → 16
Raw acceleration window
3 × 1250 samples
1D-CNN encoder → 16
Concatenate (96) → Linear → 64
One graph per trial:
6 zones × 73 windows × 64
(c) Physics-informed zone graph
Kinematic edge: w = 1 for neighbouring zones
Thermal edge: $w_{ij} = \exp\left(-\frac{d_{ij}^2}{4\,\alpha\,\Delta t_{ij}}\right)$
$\Delta t_{ij} = d_{ij}/v_f, \quad \alpha = k/\rho c = 4.9\times10^{-5}\ \mathrm{m^2/s}$
Z1
Z2
Z3
Z4
Z5
Z6
kinematic (tool path)
thermal (heat kernel)
nodes = zones; line width = edge weight
(d) Spatio-temporal block
× 2
Gated temporal convolution
kernel 3, along the 73 windows
Residual + LayerNorm
Edge-aware graph attention
4 heads, scores use w_{ij}
(ST-GCN: graph convolution)
$e_{ij} = \mathrm{LeakyReLU}(a_l^\top W h_i + a_r^\top W h_j + b^\top w_{ij})$
$\alpha_{ij} = \mathrm{softmax}_j(e_{ij})$ (space)
gated conv over windows (time)
(e) Outputs per zone
Mean over windows
zone embedding (64)
Chatter
probability
(sigmoid)
Ra (μm)*
* trained on material-mean Ra
$\mathcal{L} = \mathrm{BCE}_w + \lambda\,\mathrm{MSE}$
class-weighted BCE, λ = 1